PS5 Q3
Quitting Behavior: Logit Model

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$$logit(P(quit = 1)) = \beta_0 + \beta_1 sex + \beta_2 dex + \beta_3 lex + \beta_4 lex^2$$

where $quit = 1$ if the worker quit within the first 6 months after employed, and is 0 otherwise.

Table 1: Logit Model Estimation Results

Variable	**Coefficient**	**(Std. Err.)**
sex	0.479*	(0.186)
dex	-0.094**	(0.014)
lex	-0.539	(0.648)
lex2	0.021	(0.026)
Intercept	6.220	(4.105)

- Covariates sex and dex are statistically significant while intercept term, lex and lex^2 are not significant.
- Positive sign of coefficient for sex implies that male workers are more likely to quit their job within the first 6 months after employed.
- Negative sign of coefficient for dexterity leads us to expect that employees with higher level of dexterity are less likely to quit.
- Although not statistically significant, negative sign of lex shows some evidence for a positive effect of education on reducing the quitting rate.
- Consider the marginal effect of lex on quitting behaviour:

$$\frac{\partial Pr(quit = 1)}{\partial lex} = \frac{exp(x\beta)}{[1 + exp(x\beta)]^2} \cdot (\beta_3 + 2\beta_4 lex)$$

As $\hat{\beta}_3 < 0$, an estimated positive sign of $\hat{\beta}_4$ indicates that the shape of the education effect is likely to be an U-shape.

Another attempt is to extend the model by adding cross-product terms $sex \cdot lex$ and $sex \cdot lex^2$, which enables us to capture information of interaction between gender and education:

$$logit(p) = \beta_0 + \beta_1 sex + \beta_2 dex + \beta_3 lex + \beta_4 lex^2 + \beta_5 sex \cdot lex + \beta_6 sex \cdot lex^2$$

Table 2: Extended Logit Model Estimation Results

Variable	**Coefficient**	**(Std. Err.)**
sex	8.268	(9.394)
dex	-0.095**	(0.014)
lex	0.768	(1.279)
lex2	-0.039	(0.055)
sl	-1.494	(1.555)
sl2	0.070	(0.065)
Intercept	-0.709	(7.419)

- Note that now only dex is significant and signs of lex and lex^2 are reversed comparing with the previous model and insignificant.
- Moreover, the insignificant estimation of $sex \cdot lex$ and $sex \cdot lex^2$ hardly provides us some convincing evidence of gender effect.

Figure 1: Effect of Education on Quitting (dex=pool mean)

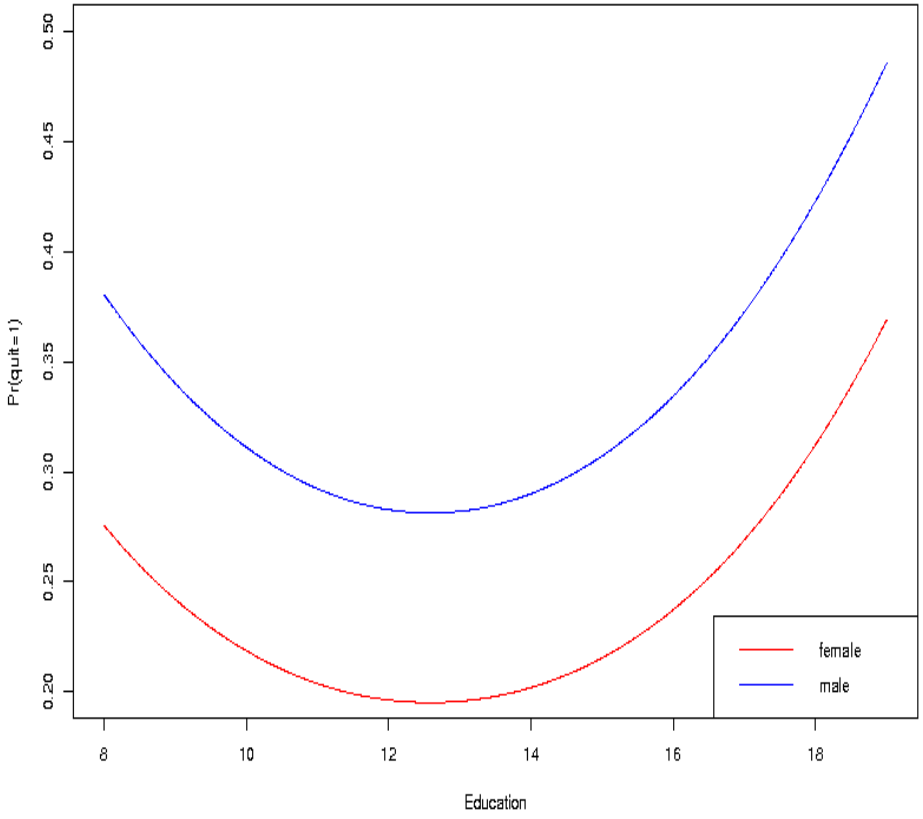
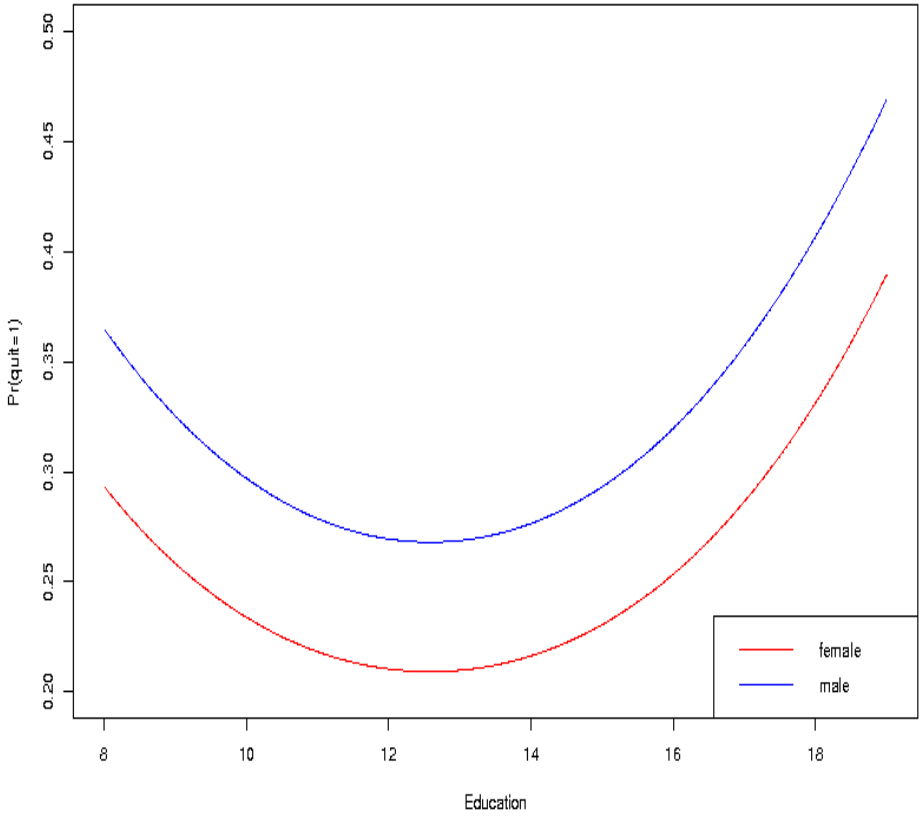
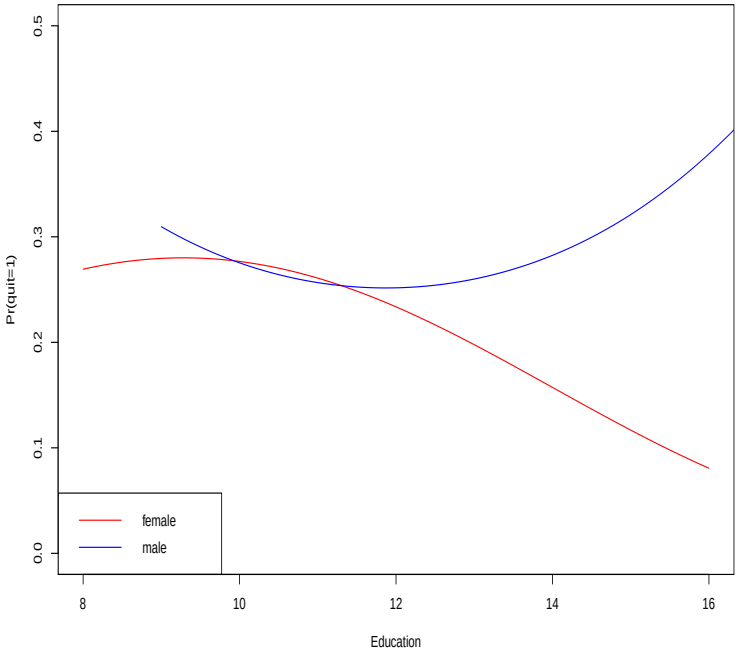


Figure 2: Effect of Education on Quitting (dex=gender mean)



However, when we do the above logit estimation for each gender separately, we find that years of education has different impact patterns on probability of quit=1 for female and male workers.

Figure 3: Effect of Education on Quitting (Group Estimation)



- As shown in Figure 3, male group has an expected U shape, while surprisingly, female group has an inversed-U shape. This implies that for $edu < 10$, female workers with higher education level are more likely to quit their job within the first 6 month of employment while for $edu > 10$, higher education decreases the intention to quit.

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- However, the sample size is relatively small for $edu < 10$ in female group (only 11 samples). Therefore, it is hard to evaluate the accuracy of estimation for the small sample sub-area.
- **To summarize**, with the same education level, female workers are less likely to quit their job than male workers. However the pattern of education effect on probability of quitting is not clear within the female group while for the male group there exhibits a U-shape of education impact.

Another important problem is to justify the logit specification. Follow Pregibon(1980), we extend model (8) as follows:

$$logit(p) = \beta_0 + \beta_1 sex + \beta_2 dex + \beta_3 lex + \beta_4 lex^2 + \gamma_1 g_0^{\alpha}(p) + \gamma_2 g_0^{\delta}(p)$$

Variables $g_0^{\alpha}(p)$, $g_0^{\delta}(p)$ are constructed from a preliminary logistic regression with

$$\begin{aligned} g_0^{\alpha}(p) &= \frac{1}{2}[log^2(p) - log^2(1-p)] \\ g_0^{\delta}(p) &= -\frac{1}{2}[log^2(p) + log^2(1-p)] \end{aligned}$$

According to Pregibon(1980), LM tests of significance of g^{α}, g^{δ} can be used to justify our logit specification. The $\chi^2(2)$ statistic is 5.66 with p-value 0.0589. Thus we accept the null hypothesis at 5% significance level. Therefore our logit model specification is a good representation of the data.

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- Hire female workers rather than male workers. Logit estimations shows significant positive estimation of sex which implies that male workers are more likely to quit their job.

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- For male and female workers who have the same level of education, hiring the female worker is better.
- Among male workers, the ones with 12 - 13 years of education are preferred because they have the smallest probability of quitting.
- To hire female workers, education should not be taken as an indicator of probability of quitting. The effect of education on $Pr(quit = 1)$ is not clear for female group.